Towards Independent Color Space Selection for Human Skin Detection

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Abstract. Skin color detection plays an important role in video based applications. Without considering the selection of suitable color space, a novel skin color detection method is proposed based on the flexible neural tree, which can identify the important components of color spaces automatically. With large training data sets, our method builds a flexible neural tree structure and optimizes its parameters using Genetic Programming and Particle Swarm Optimization algorithms. In experiments, features comprised of all channels extracted from RGB, YCbCr and HSV color spaces are used for the constructing and evaluating of the novel skin color model, in which six most important components, i.e., R, G, B, Y, Cr and S are selected for testing. Furthermore, our method achieves higher accuracy and lower false positive rate than state of the art methods on Compaq and ECU data set.

Keywords: skin detection, color space, flexible neural tree, skin classification.

1 Introduction

Skin color is a powerful and robust cue for face detection, localization and tracking. Other real time applications, like hand gesture analysis, can also benefit from automatic detection of skin color. Human skin has very consistent colors distinct from the colors of many other objects, which leads a number of pixel-based skin detection methods arising in the past few years.

The objective of pixel-wise skin detection is to build a decision rule that classifies each pixel as skin or non-skin individually. The choice of suitable color space and skin color classification techniques are two important issues in skin color detection, and a few papers have presented comprehensive studies[1,2,3]. Ref. [4] presented proof and experiments to show that the separability of the skin and non-skin classes is independent of the color space chosen. However, comparative studies and investigations of the effect of color space choice[5,6] have shown that the choice of color space not only affect the performance of skin detection, but also determines how effectively we can model the skin color distribution. As another issue, skin classification techniques can be roughly divided into three steams: fixed threshold methods, parametric methods and non-parametric methods. Mixtures of Gaussian model and elliptic boundary method[2] model the skin distribution with parameters, which have better generalization than fixed threshold methods; Bayesian classifier[7] is one of the non-parametric methods, which

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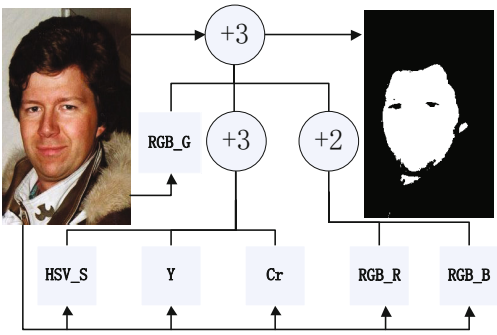


Fig. 1. The skin color model based on flexible neural tree. The model allows over-layer connections, can automatically select the input features, give higher accuracy and lower false positive rate for pixel-wise skin color detection.

usually need many storage spaces. Algorithms in machine learning, such as Mutilayer Perceptron(MLP), have also been investigated for pixel-wise skin color detection[3].

Although existing methods have good performance on skin color detection, there are problems that are not concluded: (1) The optimal color space for skin color detection is not determined. Performance of skin color detection is strongly affected by the choice of color space. A suitable color space should be determined before modeling the skin color distribution, in which exhaustive experiments would be involved; (2) Methods with higher accuracy also have higher false positive rate(FPR). In this paper, a novel pixel-wise skin color detection methods is proposed based on the flexible neural tree(FNT)[8,9](see Fig. 1), which has two contributions: (1) FNT based skin color model allows input variable's selection. The choice of color space is not a problem here, because the model can automatically select the suitable and important color channels from various color representations; (2) In the training stage, individuals with simplicity structure and similar accuracy model will be selected and reserved, which makes our method giving higher accuracy and lower FPR compared with state of the art methods.

2 FNT Based Skin Color Model

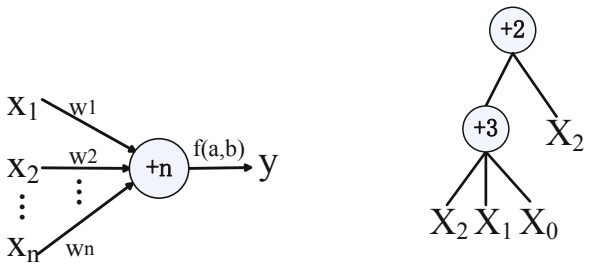
Addressing the problem of existing mentioned above, FNT based skin color model is proposed. For skin color detection, the FNT model enable automatic components selection of color spaces and better performance of skin pixel classification.

2.1 Flexible Neural Tree

Flexible neural tree is a special multi-layer feed forward network[8], allows over-layer connections, input variables selection and different activation functions for different nodes. A flexible neural tree can be generated by function set F and terminal instruction set T, which are described as follows:

$$S = F \cup T = \{+_2, +_3, ..., +_N\} \cup \{x_1, ..., x_n\} \tag{1}$$

Function set F contains multi-category non-leaf nodes named flexible neurons. $+_i(i = 1, 2, 3, ..., N)$ means instructions of non-leaf nodes that taking i inputs. The terminal instruction set T with elements $x_1, x_2, ..., x_n$ are instructions of leaf nodes taking no other arguments. A general flexible neuron operator is illustrated in Fig. 2(a).



(a) Flexible neuron operator (b) FNT instance

Fig. 2. The flexible neuron and an instance of flexible neural tree

Given predefined model structure M_i, which has function set F_i and terminal instruction set T_i, the hierarchical structure and parameters of FNT model can be evolved and optimized using Genetic Programming algorithm(GP) [10] and Particle Swarm optimization algorithm(PSO)[11], respectively. Fig. 2(b) shows a trained FNT model with function set $F = \{+_2, +_3\}$ and terminal instruction set $T = \{x_1, x_2, x_3\}$ (similar to $M1$ model in Table. 1).

In the stage of FNT modeling, random structures and corresponding parameters are evolved and optimized recursively, until a satisfactory solution is found or a time limit is reached. If Gaussian activation function(Equ. 2) is used, the two adjustable parameters a_i and b_i will be randomly created as flexible activation function parameters.

$$f(a_i, b_i, x) = e^{-((x-a_i)/b_i)^2} \tag{2}$$

The overall output of FNT model is the total excitation of the root flexible neuron(i.e., $+_2$ node in Fig. 2(b)).The excitation of $+_n$ is $net_n = \sum_{j=1}^{n} w_j \cdot x_j$, where $x_j(j = 1, 2, ..., n)$ are the input features to node $+_n$. Combined with Equation. 2, the output of the node $+_n$ is calculated by,

$$out_n = f(a_n, b_n, net_n) = e^{-((net_n - a_n)/b_n)^2} \tag{3}$$

The overall output of FNT model can be computed from left to right by depth-first method recursively.

2.2 Structure Optimization with GP

There are three important issues for FNT based skin color modeling: optimization of structure, optimization of parameters, and general learning procedure. In initialization,

feature f comprised of x_i is viewed as input, written as $f = \{x_1, x_2, ..., x_n\}$, where n is the number of samples. x_i denotes the vector $(x_{i1}, x_{i2}, ..., x_{ij})^T$, where j is number of channels of all color spaces. To find the optimal model, the root mean squared error(RMSE),

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \hat{x_i})}{N}} \tag{4}$$

is employed as fitness function, where the x_i denotes the actual skin data, $\hat{x_i}$ is the output of the skin model. The procedure of structure optimization with GP is illustrated in Algorithm. 1.

Algorithm 1. Structure optimization with GP

```
Data: M=Population Size; N=Maximum of Run
Result: Optimal Structure
Run = 0;
for i=1 to N do
    Gen = 0;
    Generate initial population randomly;
    while not reach terminate condition for Run do
        Evaluate fitness of each individual;
        for j=1 to M do
            Op = select genetic Operator;
            switch Op do
                case neural tree mutation operators
                    do as Ref. [8];
                otherwise
                    do as standard GP[10];
                endsw
            endsw
            Gen = Gen+1;
        end
        designate result for run;
        Run = Run + 1;
    end
end
```

Optimization using Genetic Programming starts with an initial population composed of F set and T set(see Tab. 1). After execution of each individual program in the population, the fitness function provided in Equ. 4, will be used to measure how well it performs the task of each individual program. Genetic operations, such as crossover, mutation and reproduction, are applied to individuals that are probabilistically selected from the population based on fitness. In the flexible neural tree structure, the leaf node can be a neural sub-tree or a terminal neuron. The crossover operator would select two neural sub-trees randomly and select one non-terminal node in the hidden layer for each neural tree randomly, and then swap the selected subtree with a pre-defined probability. Additional neural tree mutation variation operators are defined in Ref. [8].

After the genetic operations are performed on the current population, the new generation replaces the current population. The run of genetic programming terminates when the termination criterion is satisfied.

2.3 Parameter Optimization with PSO

In the construction of FNT based skin model via Algorithm. 1, its parameters, i.e., the weights $\{w_j\}$ and adjustable parameters a_i and b_i, will be optimized using PSO algorithm[11]. Let $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}$ be the fitness function, and $\mathbf{S}$ be the number of particles in the swarm, each having a position $\mathbf{x_i} \in \mathbb{R}^n$ in the searching space and a velocity $\mathbf{v_i} \in \mathbb{R}^n$. The goal is to find a solution **a** for which $f(a) \leq f(b)$ for all **b** in the search space. The details of parameter optimization are depicted in Algorithm. 2.

Algorithm 2. Parameter optimization using PSO algorithm

```
Data: w_i, a_i and b_i
Result: Best solution g
foreach particle i = 1, ...S do
    Initialize the particle's position x_i with w_i ;
    Initialize the particle's best known position: p_i ← x_i ;
    if f(p_i) < f(g) then
        update the swarm's best known position: g ← p_i ;
    end
    Initialize the particle's velocity: v_i ;
end
repeat
    foreach particle i = 1, ..., S do
        foreach dimension d = 1, ..., n do
            Pick random numbers:r_p, r_g ~ U(0, 1) ;
            Update the particle's velocity as Equation.5 ;
        end
        Update the particle's position as Equation.6 ;
        if f(x_i) < f(p_i) then
            Update the particle's best known position: p_i ← x_i;
            if f(P_i) < f(p_g) then
                Update the swarm's best known position: g ← p_i ;
            end
        end
    end
until Maximum iterations or minimum error criteria;
```

The population of particles $\{w_j\}$, a_i and b_i is randomly generated initially. Each particle keeps track of its own best position in vector $\mathbf{p_i}$. The best position among all the particles is kept track of as $\mathbf{p_g}$. At each step t, by using the individual best position, $\mathbf{p_i}$, and the global best position, $\mathbf{p_g(t)}$, a new velocity for particle i is updated by,

$$\mathbf{v_i}(t+1) = \mathbf{v_i}(t) + c_1\phi_1(\mathbf{p_i}(t) - \mathbf{x_i}(\mathbf{t})) + c_2\phi_2(\mathbf{p_g}(t) - \mathbf{x_i}(\mathbf{t})) \tag{5}$$

where c_1 and c_2 are parameters selected by the practitioner. ϕ_1 and ϕ_2 are uniformly distributed random numbers in [0,1]. Based on the updated velocities, each particle changes its position according to the following equation:

$$\mathbf{x_i}(t+1) = \mathbf{x_i}(t) + \mathbf{v_i}(t+1) \tag{6}$$

2.4 General Learning Algorithm

With predefined F set and T set, a FNT based skin color model M can be created and evolved via a general learning algorithm, which is combined the structure optimization(Algorithm. 1) and parameter optimization(Algorithm. 2).

Algorithm 3. General Learning algorithm

Require: Function set F and features f(as T set)
Ensure: Optimal model M

1: Initialization. Create an initial population randomly, set initial values of parameters used in GP and PSO algorithms;
2: Structure optimization. Optimal structure will be achieved by the natural tree variation operators as described in Section 2.2 with GP algorithm;
3: If a better structure is found, then go to step (4), otherwise go to step (2);
4: Parameter optimization. The architecture of FNT is fixed in this step, the parameters (weights and adjustable parameters) encoded in the best tree formulate a particle;
5: If the maximum number of the local search is reached, or no better parameter vector is found for a significantly long time then go to step (6); otherwise go to step (4);
6: If satisfactory solution is found, then the algorithm is stopped; otherwise go to step (2);
7: **return** Optimal FNT based skin model.

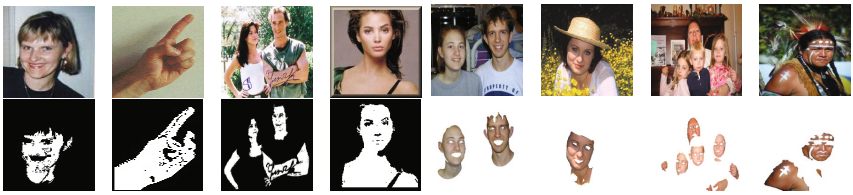
For FNT based skin color model, the nature of model construction procedure allows the model to identify important input features that is computationally efficient and effective. The mechanisms of feature selection in the model constructing procedure is illustrated as follows:

1. The components in $x_i f$ are selected to formulate the FNT based skin model with same probabilities;
2. The components which have more contributions to the objective function will be enhanced and have high opportunity to survive in the next generation by a evolutionary procedure;
3. The evolutionary operators i.e., crossover and mutation, provide a input selection method by which the FNT should select appropriate variables automatically;
4. return selected features.

3 Experiments

In experiments, Compaq[7] and ECU[6] data sets are used for the training and testing of FNT based skin color detection method. The two data sets consist of original color

images and ground-truth images, which ensure diversity in terms of the background, lighting conditions and skin types. We randomly select 2,000 images for training and testing from each data set.



(a) Sample images in Compaq set (b) Sample images in ECU set

Fig. 3. Example images in data set. Skin images with ground truth of Compaq data set are shown in (a), and (b) shows color images with corresponding skin segmented regions in ECU data set.

Some example images in Compaq and ECU data set are shown in Fig. 3. Skin pixels are taken from skin segmented images in Compaq and ECU sets. Non-skin pixels are taken from non-skin images in Compaq and the complements of skin segmented images in ECU, respectively. In testing, the skin segmented images are compared pixel-wise with the corresponding ground truth images. No extra steps are used. The performance of our methods and other methods are measured in terms of true positive rate(TPR), false detection rate(FPR) and accuracy(Acc).

For the purpose of pixel based skin color detection, three most popular color spaces RGB, YCbCr and HSV are selected for our experiments. For each pixel in each color space, a feature vector can be created by combined all the three channels in each color space. The amount of data for the training and testing of our methods is over 17GB in Compaq data set and ECU data set. We construct five FNT based skin color models with a different number of features and instruction set, as shown in Table.1. Three types of feature vector are constructed: (1) 3f: combined R, G and B channels of RGB color space; (2) 6f: Y, Cb, and Cr channels of YCbCr color space are added to 3f; (3) 9f: H, S, V of HSV color space are added to 6f.

Table 1. Different FNT based Skin Color Models

Model	F-Set	T-Set	Features
M1	$\{+_2, +_3\}$	$\{x_1, x_2, x_3\}$	3f
M2	$\{+_2, +_3\}$	$\{x_1, x_2, ..., x_9\}$	9f
M3	$\{+_2, +_3, ...+_6\}$	$\{x_1, x_2, x_3\}$	3f
M4	$\{+_2, +_3, ...+_6\}$	$\{x_1, x_2, ..., x_6\}$	6f
M5	$\{+_2, +_3, ...+_6\}$	$\{x_1, x_2, ..., x_9\}$	9f

With a subset data(30% of Compaq data set), these five models were trained and tested. As shown in Table.2, accuracy of each FNT based skin color model is greater

than 93% and FPR is lower than 6%. $M2$ model has the better performance in terms of trade-off between accuracy and FPR, which will be used in the comparison with state of the art methods. Fig.1 shows the framework of FNT based skin color model with $M2$ model after being trained using Compaq data set.

Table 2. Comparison of Five FNT based Skin Color Models

	M1	M2	M3	M4	M5
TPR	96.86%	94.97%	96.37%	93.96%	93.39%
FPR	5.70%	2.60%	5.99%	4.06%	2.32%
Acc	95.59%	96.18%	95.19%	94.95%	95.53%

Histogram based Bayes Classifier(HB), Gaussian Mixture model(GMM) , Elliptic Boundary Model(EB) and Multi-layer Perceptron Classifier(MLP) are selected for experiments, which have good performance in reported works. The parameters of HB and MLP are trained using our training data, and GMM, EB use parameters published by Jones[7] and Lee[12]. For the MLP classifier, we trained the network using Levenberg-Marquardt algorithm and Back Propagation algorithm. Different network size and activation functions were investigated, and the best network was reserved.

Table 3. Comparison Of Skin Color Classifiers

	HB	GMM	EB	MLP	Proposed
TPR^1	91.47%	89.12%	84.68%	90.12%	90.87%
FPR^1	10.19%	10.80%	19.52%	12.24%	6.16%
Acc^1	90.17%	89.18%	81.38%	89.57%	92.27%
TPR^2	83.27%	80.83%	90.43%	88.91%	86.78%
FPR^2	10.29%	15.75%	24.05%	13.70%	9.71%
Acc^2	88.47%	83.59%	78.73%	87.63%	90.79%

[1] for Compaq data set.
[2] for ECU data set.

As shown in Table.3, without extra pre-processing and post-processing, our method shows state of the art performance in accuracy and TPR, which achieves an TPR of 90.87% and accuracy of 92.27% for Compaq data set, TPR of 86.78% and accuracy of 90.79% for ECU data set. Furthermore, the FPR of 6.16% for Compaq data set and 9.71% for ECU data set is lower than other methods, which will provide detailed skin color detection. Because the training data and testing data are randomly selected for experiments, the results of other methods do not re-appear as reported in previous works, but the same data sets used for training and testing is guaranteed.

Some skin color detection results are shown in Fig.4. The first row (a) in Fig.4 shows the original color images. The ground-truth image and results of skin color classifiers

are illustrated in row (b) through (g). Compared with he ground-truth image, our method classifies the skin pixel and non-skin pixel more correctly with lower false positive detection than other methods.

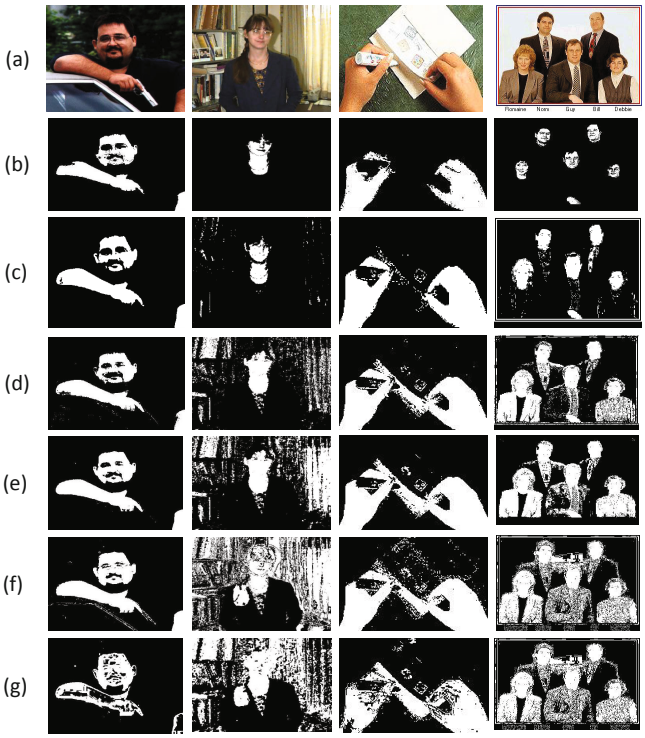


Fig. 4. Sample results of skin classifiers. The original image and its ground-truth is shown in row(a) and (b). The detection results of our method(c), HB(d), GMM(e), MLP(f) and EB(g) are depicted in bottom rows.

4 Conclusions

In this paper, we have introduced a novel human skin detection method based on the flexible neural tree for pixel-wise skin color detection. Without considering the selection of suitable color spaces, the proposed method selects important components of color spaces automatically. Our method presents a solution of the choice of color spaces. Furthermore, FNT based skin detection method achieves higher accuracy and lower false positive rate compared with state of the art methods. In the training stage, the individuals which have simple structures and similar model accuracy will be selected, which makes the structure of our model is generally simpler. The optimized hybrid tree structure guarantees state of the art performance in skin pixel classification but has shortage of time consuming for the optimization of structure and parameters. It is not a problem with high-performance computers and parallel algorithm, which will be the next part of our work.

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